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The Strength and Fatigue Analysis of Pipe Riser Reinforcement for Cylindrical FPSO

Jinghui Duan and Boqian Yu, Offshore Engineering Technology Center of CCS, China

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Abstract

The cylindrical FPSO plays an important role in the oilfield development. The strength analysis of the riser connection reinforcement structure faces new technical challenges, and it is necessary to verify the effectiveness and reliability of the connection reinforcement structure. By establishing a finite element model, the force characteristics of the riser connection reinforcement structure are analyzed. According to the specific loads borne by the local structure, typical load cases are selected for calculation to obtain the stress results of the local structure. The result analysis reveals that attention should be paid to the stress concentration phenomenon in the local structure of the riser connection reinforcement. Local stiffeners should be set to keep the stress value within a reasonable range. The fatigue of the pipe riser reinforced structure was analyzed by using the deterministic method by selecting different positions, and it is verified that the fatigue of the pipe riser reinforced structure could meet the expected service life of the structure. During the actual operation period, the changes of the structure should be monitored, and the inspection should be strengthened to ensure the integrity of the local structure of the riser connection reinforcement. The safety of the local structure is crucial for the safe operation of the cylindrical FPSO. Methods such as finite element model analysis can be adopted to complete the strength verification of the structure.

Key words: strength analysis, fatigue analysis, pipe riser reinforcement, cylindrical FPSO

Introduction

FPSO (Floating Production Storage and Offloading) is a highly sophisticated offshore floating facility that serves as a comprehensive solution for hydrocarbon production, storage, and offloading. This integrated system plays a pivotal role in offshore oil and gas operations, particularly in remote or deepwater fields where fixed infrastructure is impractical. The FPSO primarily consists of two major components: upper module (process module) and lower hull (storage and stability module). The upper module houses the processing systems that separate and treat the multiphase mixture (oil, gas, and water) delivered from subsea pipelines. It includes key equipment such as separators, dehydration units, gas compressors, and water treatment systems to ensure the extracted crude meets export standards. The lower hull functions as a massive floating storage tank, designed to safely hold processed crude oil and associated natural gas.

Once the storage threshold is reached, the crude is offloaded via a dynamic offloading system (e.g., turret or hose systems) to shuttle tankers for transportation to refineries. The FPSO concept has gained widespread adoption among major oil and gas companies due to its flexibility, cost-effectiveness, and adaptability to varying field conditions. Over decades of industry development, specialized FPSOs have been engineered to suit different operational environments, leading to two primary structural classifications: ship-shaped FPSOs and cylindrical FPSOs.

The turret single-point mooring system requires a large investment and has a high maintenance cost[1]. Cylindrical FPSOs can adopt multi-point mooring instead of single-point mooring. With symmetric arrangement, they have weak anisotropy and no weathervaning effect. Without a single-point system, the hull structure is relatively economical. However, due to its own structural characteristics, the natural period of the heave motion of the cylindrical FPSO is close to the spectral peak period of most sea conditions. Therefore, the heave motion performance of the cylindrical FPSO is poor, and it is necessary to add a heave damping plate at the bottom to significantly reduce the peak value of the heave RAO.

Marine risers are key equipment in the process of oil and gas extraction, connecting offshore platforms and subsea pipelines "from top to bottom"[2]. Its structural integrity under cyclic loading and harsh environmental conditions is paramount, requiring rigorous strength and fatigue verification[3].Under the long-term action of alternating loads such as sea wind, ocean currents, surges, and top tension, marine risers used for deepwater oil and gas extraction are prone to damage[4,5]. The structural behavior response of the riser can be predicted by means of the finite element method[6]. Strength verification focuses on assessing the riser's ability to withstand extreme loads without failure, encompassing burst, collapse, axial tension/compression, and combined loading scenarios (e.g., tension-bending). Industry standards like API 17B[7]and API 2RD[8] provide fundamental design guidelines and acceptance criteria, typically based on limit state design principles. Verification involves detailed finite element analysis (FEA) to model complex geometries and boundary conditions, calculating stresses and comparing them against material yield and ultimate strengths. For reinforced sections, such as thickened welds, welded-on stiffening rings (buckle arrestors, fatigue sleeves), or composite repair systems, modeling the interaction between the base pipe and reinforcement is crucial. Fatigue verification addresses damage accumulation from cyclic stresses caused by wave action, vortex-induced vibration and operational pressure fluctuations. The industry-standard approach employs the *S-N* curve methodology (stress-life) coupled with Miner's rule for cumulative damage calculation. The loading history is significant in the fatigue analysis by defining long-term environmental conditions and operational profiles. The cyclic stress ranges often require sophisticated analysis techniques like spectral analysis or time-domain simulation using global and local FEA models[9]. Stress Concentration Factors (SCFs) derived from FEA or parametric equations are essential for welded joints and reinforcements. As to the damage calculation, the appropriate S-N curve and Miner's rule should be applied to predict fatigue life[10]. Fracture mechanics analysis is increasingly used for flaw assessment and life extension of existing risers, particularly at reinforced sections or identified flaws. Reinforcement structures primarily aim to enhance fatigue performance at critical zones like the touchdown point of steel catenary risers or near connectors. Common techniques include weld profiling to improve the toe geometry, adding thick-walled sleeves, or utilizing composite materials for repair and reinforcement[11]. Strength and fatigue verification of risers and their reinforcement structures relies on a combination of stringent standards, sophisticated finite element model (FEM), and targeted experimental validation. Ongoing research focuses on refining riser predictions for complex reinforcements[12], advancing fracture mechanics applications for life assessment, developing high-performance composite repair systems[13], and improving the accuracy of long-term load prediction and cumulative damage modeling to enhance the safety and longevity of these vital offshore components. It is necessary to establish a model to study the safety of the pipe riser reinforcement structure.

The finite element model

Coordinate system

The following figure shows the coordinate system used in the FE model. The positive *X* towards platform EAST. The positive *Y* towards platform NORTH. The positive *Z* upwards from 0 at platform keel.

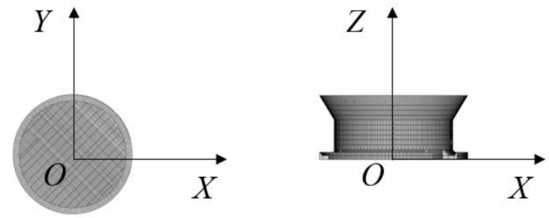

Figure 1—Coordinate System

Pipe riser reinforcement structure

The pipe riser reinforcement is located at water ballast tank through the main deck to the bottom as shown in Figure 2. The appropriate element size is adopted to meet the requirements of calculation accuracy. Generally, the element size is determined in accordance with the standard spacing (200mm × 200mm). The fine mesh size of 50mm × 50mm is used around the pipe riser structure. The riser reinforcement is the center and the model expands outward for a sufficiently long distance. All boundary displacements are fixed for *X*, *Y*, *Z*, while angular displacements are released, as shown in Figure 3. According to the design of the cylindrical FPSO, the pipe and cable interface load provides the tension of the top interface, as well as the shear and bending moments of the bottom. The stiffened panel and girder web structure are modeled with SHELL elements, while girder flange and stiffener are modeled with BEAM elements. The local model of the main deck area is shown in Figure 4 and the bottom is shown in Figure 5.

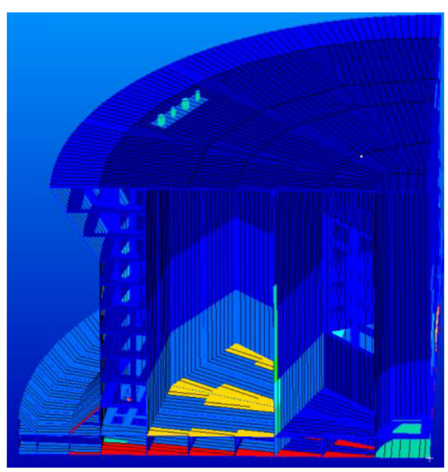

Figure 2—Model of Pipe Riser Reinforcement

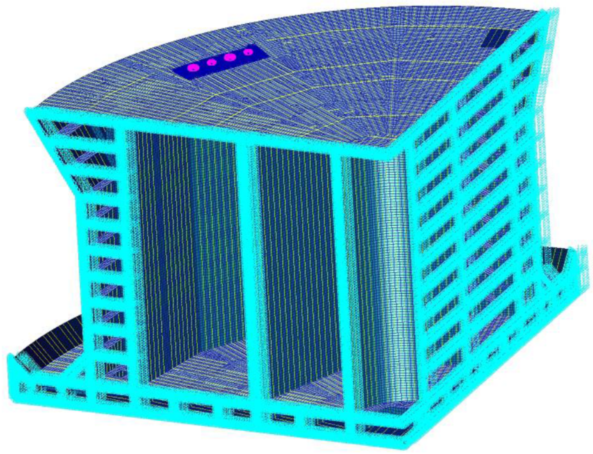

Figure 3—Boundary conditions of pipe riser reinforcement

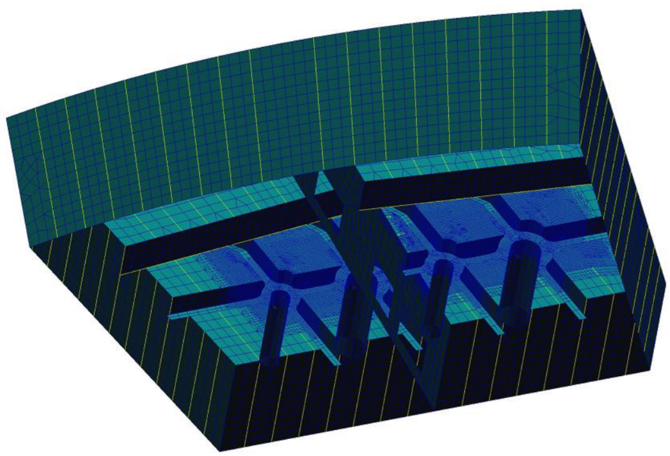

Figure 4—Local model of the main deck area.

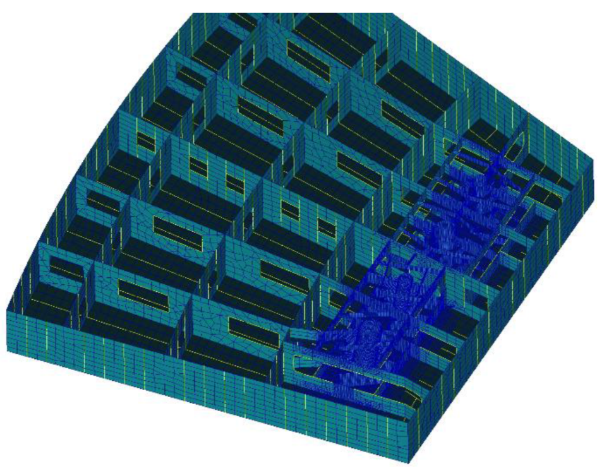

Figure 5—Local model at the bottom

3 Design load

Dead load and deck load

The self-weight of the structure is generated by utilizing specified member properties and gravity. According to the deck load plan, the 25000 Pa is applied on the deck.

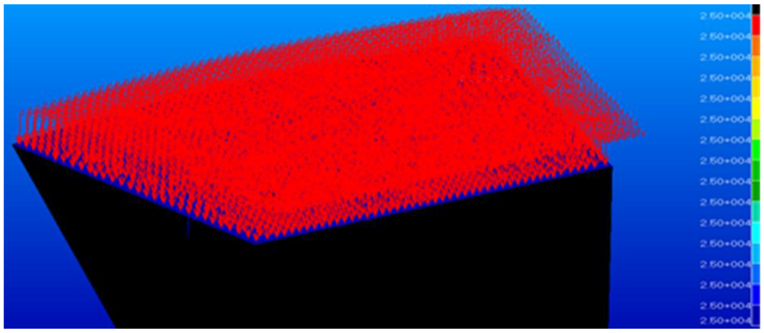

Figure 6—The deck load

Water pressure and green sea load

According to the loading conditions of different tanks, the water loads are applied in the form of surface pressure. In addition, the green sea loads on the main deck are also considered as the most critical condition. The water stopper wall is 1.2 meters around the main deck, Therefore, about 12 kPa of the green sea load is applied on the main deck in the vertical direction.

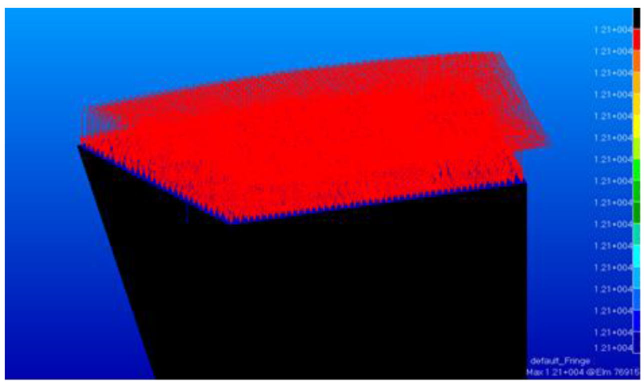

Figure 7—The green sea loads

Load cases

The pipe riser is subjected to loads at its top and bottom interfaces respectively. The vertical load on hose pipe at the top interface is about 1800 KN and the shear load acting on hose pipe is about 900 KN. The vertical load on cable pipe at the top interface is about 750 KN and the shear load acting on cable pipe is about 375 KN. There are different load cases displaying all variety situations that may happen at the bottom interface of pipe riser, as shown in the following tables.

Table 1—Load cases for cable pipes

Load cases	*Fx* (kN)	*Fy* (kN)	*Mx* (kN.m)	*My* (kN.m)
LC_001	89.72	-593.25	-1334.82	-201.87
LC_002	482.94	-356.05	-801.12	-1086.61
LC_003	593.25	89.72	201.87	-1334.82
LC_004	356.05	482.94	1086.61	-801.12
LC_005	-89.72	593.25	1334.82	201.87
LC_006	-482.94	356.05	801.12	1086.61
LC_007	-593.25	-89.72	-201.87	1334.82
LC_008	-356.05	-482.94	-1086.61	801.12

Table 2—Load cases for hose pipes

Load cases	*Fx* (kN)	*Fy* (kN)	*Mx* (kN.m)	*My* (kN.m)
LC_001	360.31	-824.73	-2474.18	-1080.94
LC_002	837.95	-328.39	-985.17	-2513.85
LC_003	824.73	360.31	1080.94	-2474.18
LC_004	328.39	837.95	2513.85	-985.17
LC_005	-360.31	824.73	2474.18	1080.94
LC_006	-837.95	328.39	985.17	2513.85
LC_007	-824.73	-360.31	-1080.94	2474.18
LC_008	-328.39	-837.95	-2513.85	985.17

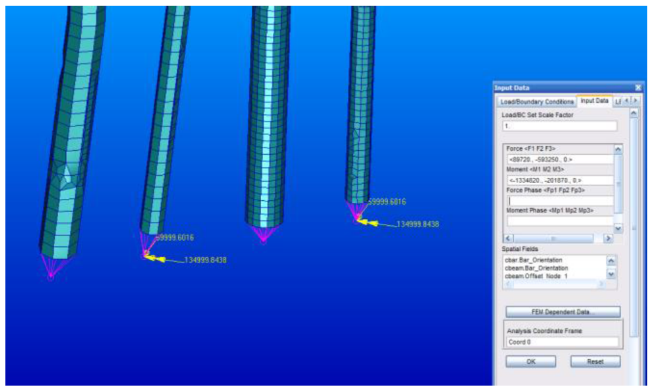

Figure 8—Shear load and bending moment acting on hose pipe at bottom interface

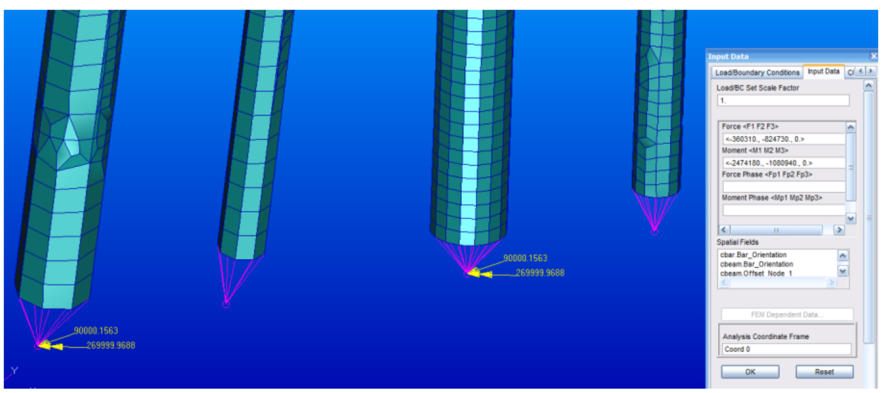

Figure 9—Shear load and bending moment acting on cable pipe at bottom interface

Strength analysis

Through the finite element model, the stress results of the cylindrical FPSO pipe riser reinforcement structure under 8 load cases are obtained. The max von-Mises stress of pipe riser at top is 178 MPa for LC001-008, as shown in Figure 10. The max von-Mises stress of the top structure around pipe riser is 139 MPa for LC001-008, as shown in Figure 11. The max von-Mises stress results of pipe rise and around structure at bottom are listed in Table 3. For the pipe riser reinforcement structure, the material grade is S355 which has been certified by the classification society, and the specified minimum yield stress of the material is 355 N/mm2 for plates, stiffeners, tubular supporting members and girders. In this analysis, the allowable stress taken as 0.8 of the material yield stress is 284 MPa. The maximum stress of the structure is 273 MPa, which is less than the allowable stress, therefore the structure of pipe riser reinforcement is safe.

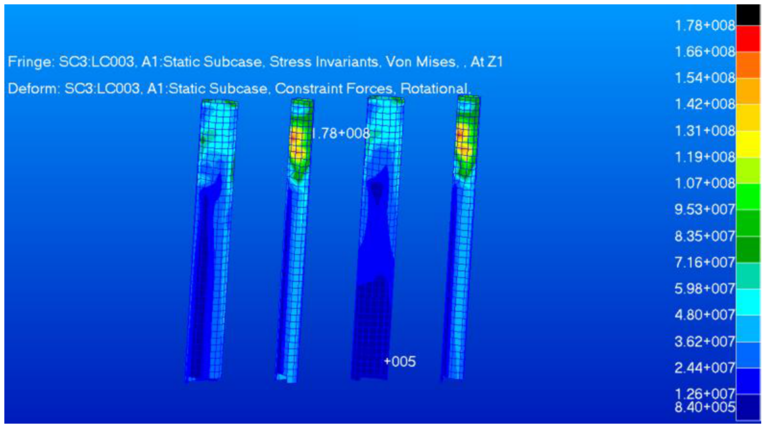

Figure 10—Von Mises stress of pipe riser at top

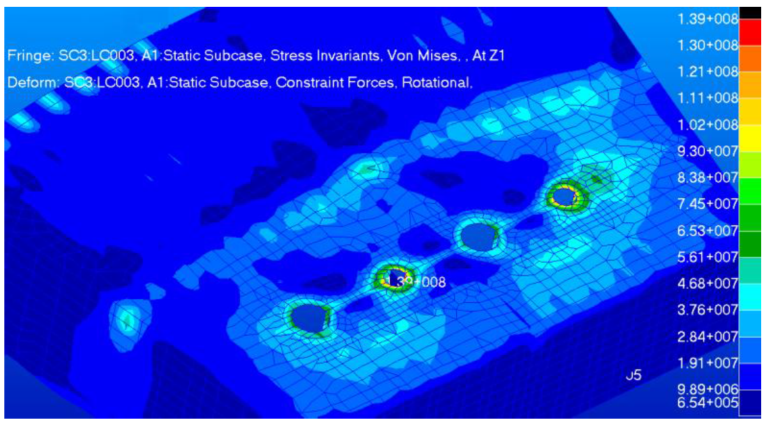

Figure 11—Von Mises stress of the top structure around pipe riser

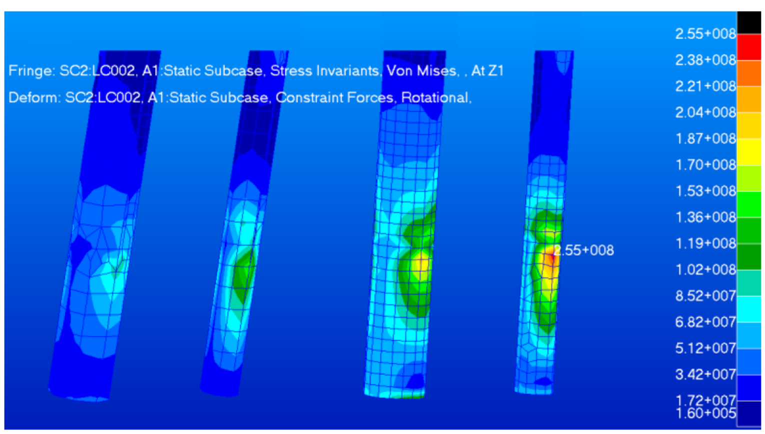

Figure 12—Von Mises stress of pipe riser at bottom

Figure 13—Von Mises stress of the bottom structure around pipe riser

Table 3—The von-Mises stress of pipe rise and around structure at bottom

Load cases	*Fx* (kN)	*Fy* (kN)	*Mx* (kN.m)	*My* (kN.m)	Pipe riser at bottom (MPa)	structure around pipe riser at bottom (MPa)
LC_001	89.72	-593.25	-1334.82	-201.87	215	142
LC_002	482.94	-356.05	-801.12	-1086.61	255	173
LC_003	593.25	89.72	201.87	-1334.82	217	136
LC_004	356.05	482.94	1086.61	-801.12	231	133
LC_005	-89.72	593.25	1334.82	201.87	216	137
LC_006	-482.94	356.05	801.12	1086.61	257	168
LC_007	-593.25	-89.72	-201.87	1334.82	218	146
LC_008	-356.05	-482.94	-1086.61	801.12	232	133

Fatigue Analysis

Loads and load cases

The input load cycle data for this fatigue calculation is sourced from the load cases and corresponding cycle counts provided by the equipment manufacturer and the load cycles are corresponding to 30 years. For the Hose Pipe, the manufacturer provided 168 sets of data. Each set includes axial force, horizontal shear force, horizontal bending moment, and the corresponding number of cycles. For the Cable Pipe: The manufacturer provided 810 sets of data. Each set similarly includes axial force, horizontal shear force, horizontal bending moment, and the corresponding number of cycles. As mentioned previously, these loads are applied separately to the top and bottom of the pipe riser reinforcement structure. Crucially, for each load set, the manufacturer provided both the maximum and minimum load values. Therefore, the fatigue calculation involves applying both the maximum and the minimum load values separately for each set. This results in:

Hose Pipe: 168 sets × 2 load values (max & min) per set = 336 load cases.

Cable Pipe: 810 sets × 2 load values (max & min) per set = 1620 load cases.

The deterministic fatigue analysis method

Due to the detailed fatigue load cycle statistics for the riser provided by the equipment manufacturer, the deterministic fatigue analysis method is adopted to verify the fatigue strength at the pipe riser reinforcement structure. Fatigue failure is one of the most common structural failure modes under cyclic loading. The deterministic fatigue analysis method is a classical engineering approach for predicting structural fatigue life. It calculates structural fatigue life based on deterministic load input parameters through mechanical analysis and fatigue theory, without considering parameter randomness and uncertainty. The principal stress ranges can be calculated by multiplying load to stress transfer function by load component.

The steps of deterministic fatigue analysis are as follows:

1. Perform Static Structural Analysis:
   - Input the statistical load data described above. Each dataset includes the maximum load (Max), minimum load (Min), and the number of occurrences (n) over the entire service life.
   - Apply the Max load and the Min load separately. Conduct static analyses to obtain the stress distribution results under both load conditions.
   - Subtract the two stress distribution results to obtain the alternating stress distribution caused by the load variation.
2. Calculate Stress Range:
   - Extract the Maximum Principal Stress (σ_{max}) and Minimum Principal Stress (σ_{min}) from the alternating stress distribution. Calculate the Stress Range ($\Delta\sigma = \sigma_{max} - \sigma_{min}$).
3. Select *S-N* Curve:
   - Select the appropriate *S-N* curve based on the specific location of the structural assessment point.
4. Calculate Cumulative Damage:
   - For each calculated stress range ($\Delta\sigma_i$), identify:
     a. n_i: The number of occurrences over the service life (from input data).
     b. N_i: The number of cycles to failure for that stress range (obtained from the *S-N* curve).
   - Calculate the damage contribution for each stress range: $D_i = n_i / N_i$.
   - Apply Miner's linear cumulative damage rule to compute the total cumulative damage at each assessment point over the entire service life: $D_{total} = \Sigma(n_i / N_i)$.
5. Calculate Fatigue Life:
   - The fatigue life (in years) at the assessment point for a service life of L years is calculated as: Fatigue Life = L / D_{total}.

The finite element calculation yields the stress distribution under the Max load for each load case and the stress distribution under the Min load for each load case.

For each element, extract the following principal stresses:

- σ_{max1}: Maximum Principal Stress under Max load.
- σ_{min1}: Minimum Principal Stress under Max load.
- σ_{max2}: Maximum Principal Stress under Min load.
- σ_{min2}: Minimum Principal Stress under Min load.

The Maximum Principal Stress Range (S_{max}) and Minimum Principal Stress Range (S_{min}) caused by the alternating load are then calculated as follows:

$$S_{max} = max\left(\sigma_{max1} - \sigma_{max2}, \sigma_{min1} - \sigma_{min2}\right)$$
$$S_{min} = min\left(\sigma_{max1} - \sigma_{max2}, \sigma_{min1} - \sigma_{min2}\right)$$

The calculated stress range are as follows:

$$\text{Stress Range } = 1.12 \times \max(|S_{max}|, |S_{min}|)$$

The *F*-class *S-N* curve for non-tubular joints is selected for the fatigue assessment of the Pipe Riser Reinforcement structure. Specifically:

- For the Top Section: The *F*-class curve for non-tubular joints in air is applied.
- For the Bottom Section: The *F*-class curve for non-tubular joints in seawater with cathodic protection is applied.

In this calculation, the number of load cycles n_i for each input load over the service life has been provided in the equipment manufacturer's load data. Therefore, it is only necessary to calculate the number of cycles to failure *Ni* for each stress range using the stress range values and the *S-N* curve. Derive the individual damage value n_i / N_i for each stress range. Compute the cumulative damage $\Sigma(n_i / N_i)$ by summing all individual damage values.

Since the input load cycle data (n_i) represents the cumulative occurrences over the 30-year design service life, the fatigue life at the assessment point is calculated as:

$$\text{Fatigue Life (years) } = 30 / D_{total} = 30 / \Sigma(n_i / N_i)$$

S-N Curve and Fatigue Design Factor

Most of fatigue assessments of offshore platforms are based on laboratory fatigue test data, *S-N* curves. Fatigue assessment of the reinforcement structures at the top and bottom sections of the riser will utilize the *F*-class *S-N* curve. Specifically, the top section is located above the waterline and the bottom Section is located below the waterline.

The applicable *S-N* curves for these regions are shown in the figure below.

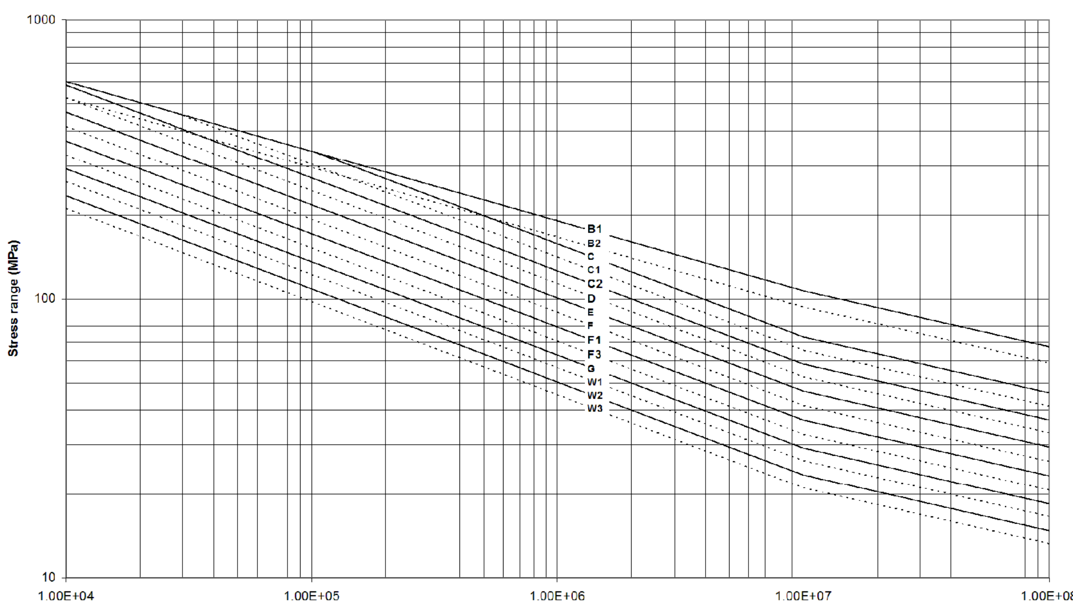

Figure 14—The *S-N* curve for non-tubular joints in air

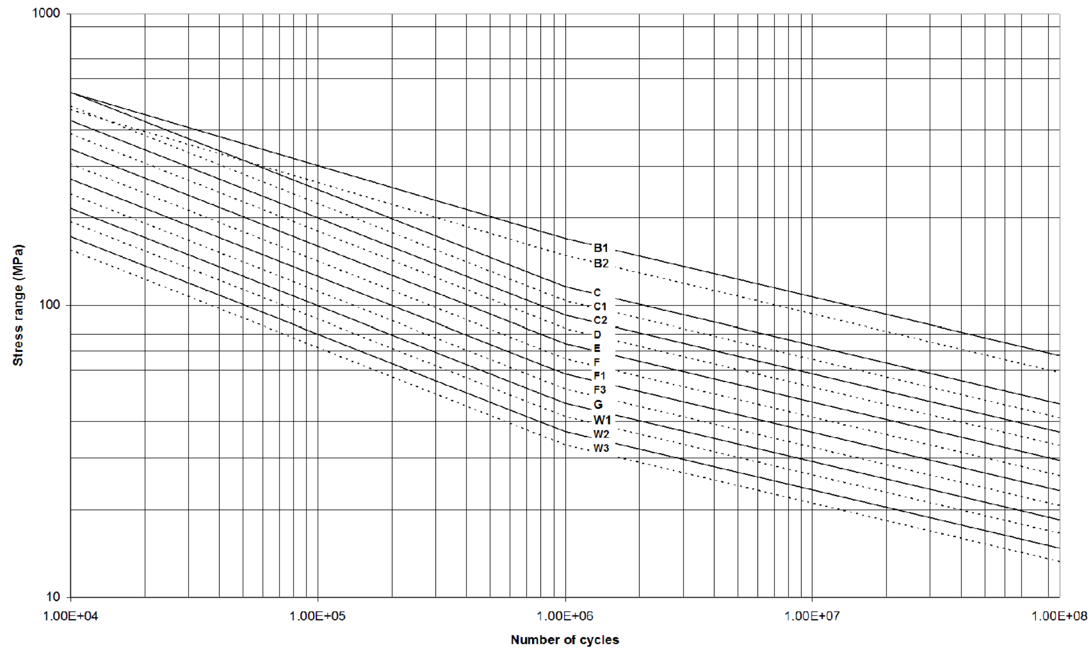

Figure 15—The *S-N* curve for non-tubular joints in seawater with effective cathodic protection

The required calculated fatigue lives are determined by safety factors together with the design life of the platform. This platform is designed for a service life of 30 years. The Design Fatigue Factors (DFF) is determined per the inspectability, repairability, redundancy, the ability to predict failure damage, as well as the consequence of failure of the structure components in question. DFF=10 is used in the analysis, and the fatigue life for these areas shall be more than 300 years. The fatigue design life of the platform during service shall be no less than 30 years, multiplied by the fatigue design factor (DFF). According to design requirements, the fatigue life of the riser connection structure is 30 × 10.0 = 300 years.

Plate Thickness Correction Factor

The fatigue strength of components is influenced by plate thickness. As plate thickness increases, the amplitude of cyclic stress decreases accordingly; thus, the effect of plate thickness on fatigue strength must be considered. Changes in plate thickness alter the local geometry and stress distribution of components, thereby affecting the fatigue life of hot spots.

When plate thickness increases, the stress range shall be multiplied by a plate thickness correction factor. The formula is as follows:

$$S_f = S\left(\frac{t}{t_{ref}}\right)^q$$

Where, S: Unadjusted stress range, t: Plate thickness, t_{ref} = 25 mm, q = 0.25.

Fatigue area

The fatigue strength verification adopts a simplified direct calculation method. Critical areas prone to fatigue failure at the top and bottom reinforcement zones of the riser are selected. Based on the computational analysis results, conclusions and recommendations are provided. The scope of this fatigue strength verification includes two Hose Pipes and two Cable Pipes. For clarity, each pipe is labeled as illustrated below. Subsequent verifications will be conducted separately for each designated area.

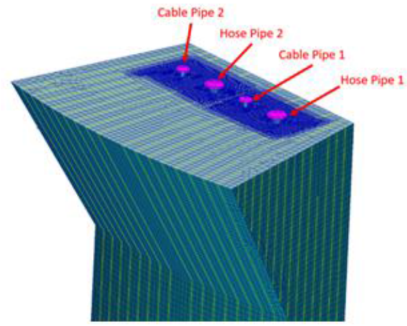

Figure 16—Nomenclature of Riser Reinforcement Structures

The fatigue life results of fatigue sensitive areas (where high Principal stress exist) in following figures are analyzed.

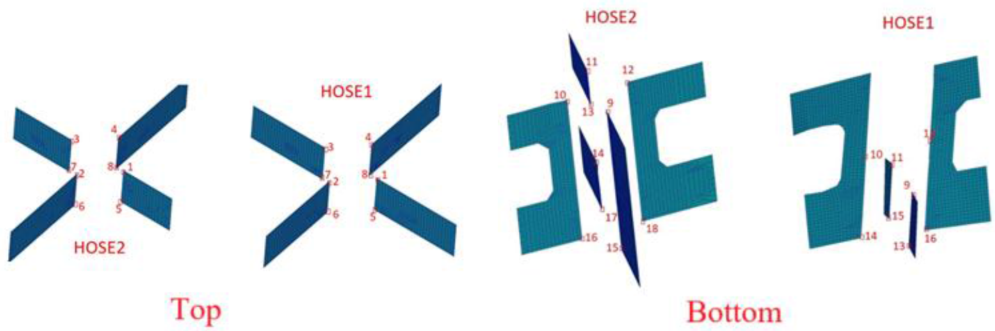

Figure 17—the plate of hose pipes at top and bottom

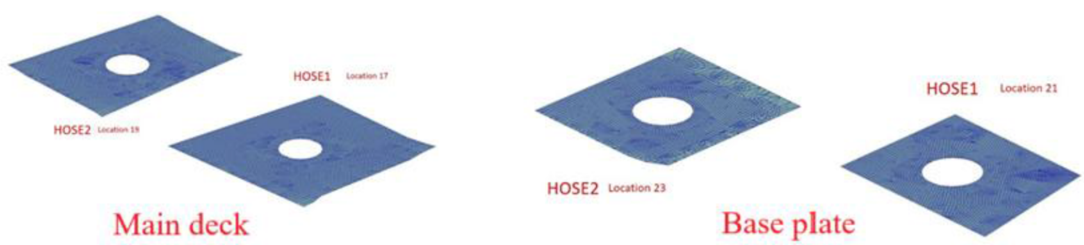

Figure 18—the main deck and base plate of hose pipes

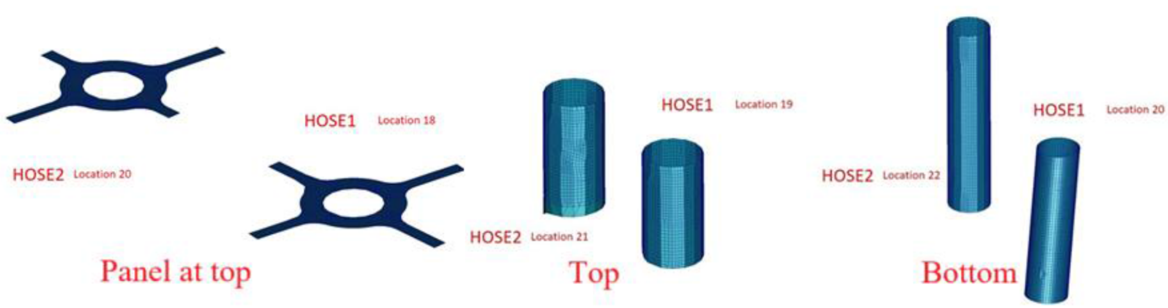

Figure 19—the panel and pipe riser reinforcement of hose pipe

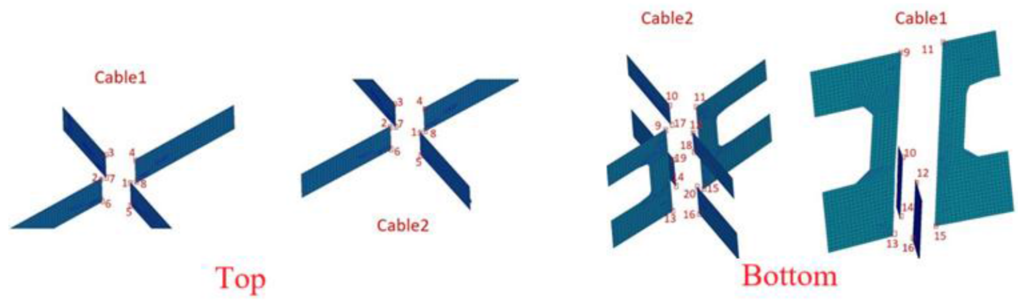

Figure 20—the plate of cable pipes at top and bottom

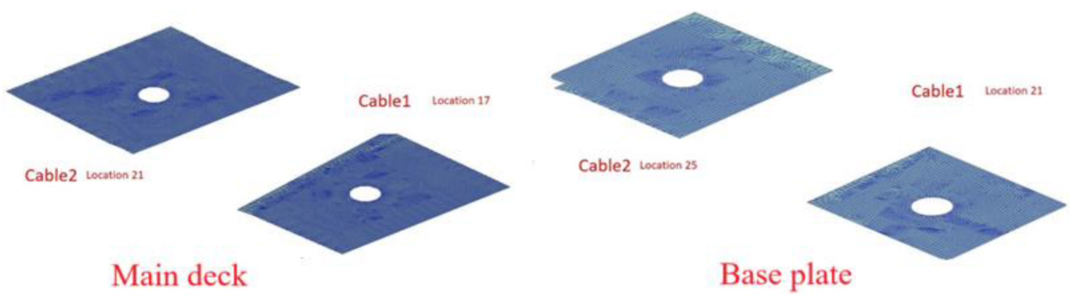

Figure 21—the main deck and base plate of cable pipes

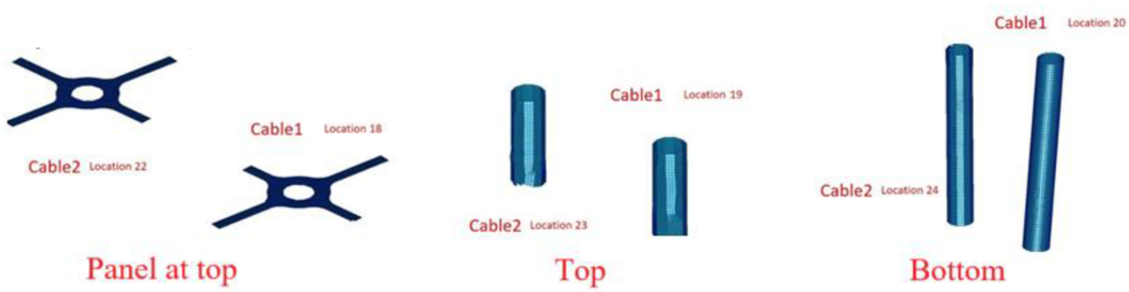

Figure 22—the panel and pipe riser reinforcement of hose pipe

Fatigue life

A series of points within the top and bottom reinforcement zones of each Hose Pipe and Cable Pipe were selected for fatigue life calculation. The fatigue life results for each verification point are summarized as presented in the table below.

Table 4—The results of fatigue life for hose pipe reinforcement

Location	Hose Pipe 1		Hose Pipe 2	
	Max Stress Range (Mpa)	Fatigue Life (year)	Max Stress Range (Mpa)	Fatigue Life (year)
1	4.661	12853947	10.036	278432
2	3.957	29170954	3.148	91507703
3	6.600	2259992	5.083	8242637
4	4.178	22208568	3.060	105300293
5	3.487	55523081	6.426	2605011
6	2.753	181542946	1.339	6567496962
7	5.183	7620560	2.574	245375744
8	3.983	28415012	1.673	2176526272
9	1.587	1262944214	1.437	8861430850
10	3.280	143958412	1.511	2932877951
11	3.495	61449070	1.097	24622069175
12	2.311	8007879754	1.631	39951246436
13	7.423	9116673	0.459	3220246500125
14	10.328	562568	1.672	9170473978
15	6.577	10824590	3.644	120494730
16	8.473	1173703	10.461	556388
17	3.479	55732387	3.190	1581485147
18	11.058	171934	10.376	602437
19	5.572	5187431	3.155	90642994
20	14.994	100855	7.162	1493802
21	1.917	1948599600	6.412	2555157
22	--	--	17.151	50366
23	--	--	2.580	805277810

Table 5—The results of fatigue life for cable pipe reinforcement

Location	Cable Pipe 1		Cable Pipe 2	
	Max Stress Range (Mpa)	Fatigue Life (year)	Max Stress Range (Mpa)	Fatigue Life (year)
1	6.143	5874945405	7.234	2510048644
2	3.646	75076018956	4.404	29867876009
3	9.133	767072584	5.309	11781915288
4	3.798	60850579810	4.116	41881936684
5	2.902	261482313307	4.650	23550893119
6	1.725	3127929006524	2.106	1240581086973
7	5.479	9571631609	3.337	123782593179
8	2.180	960222700649	3.232	145821075641
9	0.213	283475610422422	1.114	8751196115786
10	2.592	3718820619	0.964	16994401664981

Location	Cable Pipe 1		Cable Pipe 2	
	Max Stress Range (Mpa)	Fatigue Life (year)	Max Stress Range (Mpa)	Fatigue Life (year)
11	0.979	39207868996862	1.101	18814727656552
12	1.430	2998617923	0.953	42910426978755
13	5.029	125409554	4.989	1820596560
14	10.852	2659361	10.478	2426083
15	4.293	2817098846	4.992	1497879066
16	10.611	1771210	11.434	1319421
17	3.222	136181100321	1.206	5821873367217
18	7.330	2317593159	0.916	7020373148433
19	4.988	15831330445	2.757	2657859367
20	18.764	133325	0.801	1878854292028
21	4.682	64584737	3.352	118606765948
22	--	--	10.662	365592030
23	--	--	8.159	313946
24	--	--	19.088	90105
25	--	--	5.205	34431290

Conclusion

Based on the actual force characteristics of the pipe riser connection reinforcement structure, the FE model has been established under different typical load cases. The method for the strength analysis is summarized and the results of von Mises stress for the reinforcement structure cases can be obtained.

The von Mises stress of pipe riser at bottom is relatively larger and More attention should be paid during the construction of welding. The frequency of inspection on these areas should be increased during the offshore operation period.

If relevant information such as the load magnitude and cycle number during the expected service period of the structure can be obtained, the fatigue calculation of the riser connection reinforced structure can be verified by using the deterministic fatigue analysis method. In the analysis of this paper, the riser connection reinforced structure can relatively meet the service time of the structure. During the actual operation and maintenance period, the inspection of the underwater structure should be strengthened and the changes of the structure should be paid attention to. In the future, it will be necessary to improve the accuracy of the load cases and corresponding cycle counts for the fatigue calculation and more verification should be carried out in the actual engineering projects to enhance the reliability of fatigue analysis methods.

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